



Norm Space

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01

P-norm



Vector Norms

- P-norm

$$\|x\|_p = (|x_1|^p + |x_2|^p + \dots + |x_n|^p)^{\frac{1}{p}}$$

subject to $p \geq 1$

- What is the shape of $\|x\|_p = 1$?
- Properties?

Norm

Definition

- A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a norm if
1. $f(x) \geq 0$, $f(x) = 0 \iff x = 0$ (positivity)
 2. $f(\alpha x) = |\alpha|f(x)$, $\forall \alpha \in \mathbb{R}$ (homogeneity)
 3. $f(x + y) \leq f(x) + f(y)$ (triangle inequality)

02

1 - norm and 2 - norm

Vector Norms

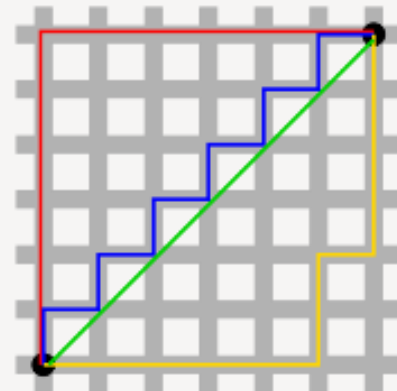
- 1-norm(l_1):

$$\|x\|_1 = (|x_1| + |x_2| + \dots + |x_n|)$$

- What is the shape of $\|x\|_1 = 1$?
- The distance between two vectors under the l_1 norm is also referred to as the **Manhattan Distance**.
- Properties?

Example

l_1 distance between $(0, 1)$ and $(1, 0)$?



Norm Derivations

□ Square of l_2


$$x = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$$

$$\|x\|_2^2 = x_1^2 + x_2^2 + \dots + x_n^2$$

$$\Rightarrow \left\{ \begin{array}{l} \frac{d\|x\|_2^2}{dx_1} = 2x_1 \\ \frac{d\|x\|_2^2}{dx_2} = 2x_2 \\ \dots \\ \frac{d\|x\|_2^2}{dx_n} = 2x_n \end{array} \right.$$

Norm Derivations

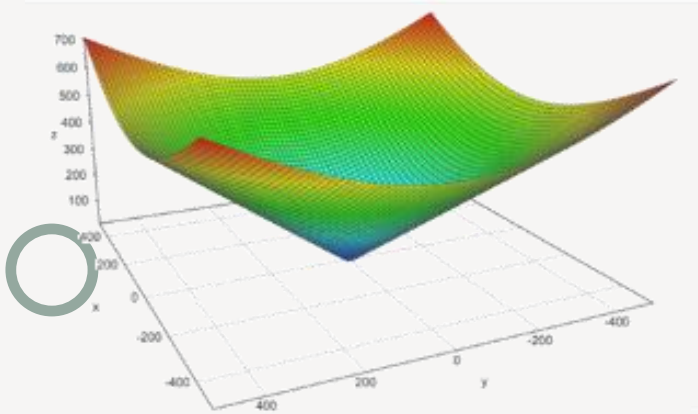
□ l_2

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \quad \|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = (x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}}$$

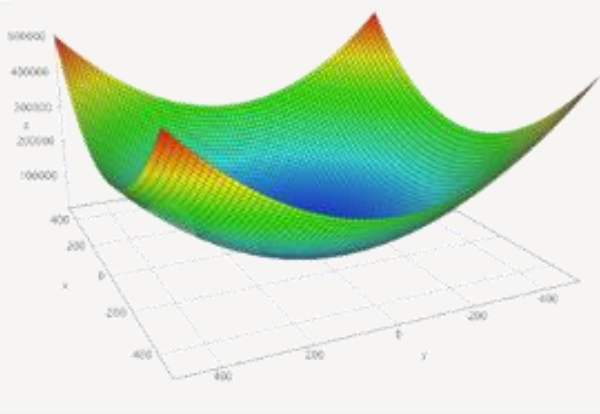

$$\begin{aligned} \frac{d\|x\|_2}{dx_1} &= \frac{1}{2} (x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}-1} \cdot \frac{d}{dx_1} (x_1^2 + x_2^2 + \dots + x_n^2) \\ &= \frac{1}{2} (x_1^2 + x_2^2 + \dots + x_n^2)^{-\frac{1}{2}} \cdot \frac{d}{dx_1} (x_1^2 + x_2^2 + \dots + x_n^2) \\ &= \frac{1}{2} \cdot \frac{1}{(x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}}} \cdot \frac{d}{dx_1} (x_1^2 + x_2^2 + \dots + x_n^2) \\ &= \frac{1}{2} \cdot \frac{1}{(x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}}} \cdot 2 \cdot x_1 \\ &= \frac{x_1}{(x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}}} \end{aligned}$$

$$\Rightarrow \begin{cases} \frac{d\|x\|_2}{dx_1} = \frac{x_1}{(x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}}} \\ \frac{d\|x\|_2}{dx_2} = \frac{x_2}{(x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}}} \\ \dots \\ \frac{d\|x\|_2}{dx_n} = \frac{x_n}{(x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}}} \end{cases}$$

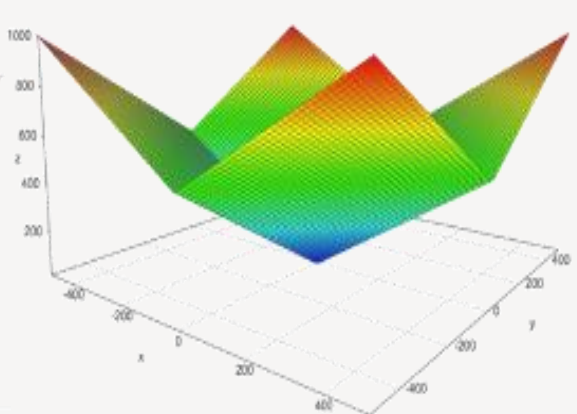
Norm Comparisons



l_2 norm

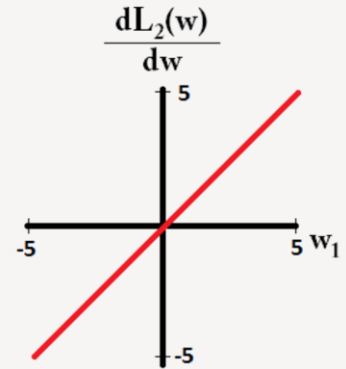
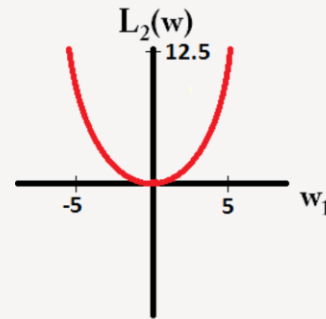
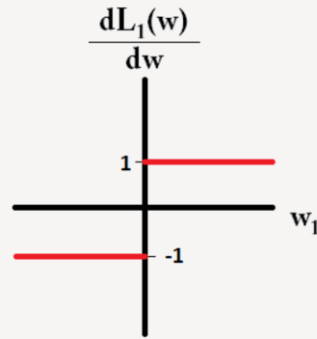
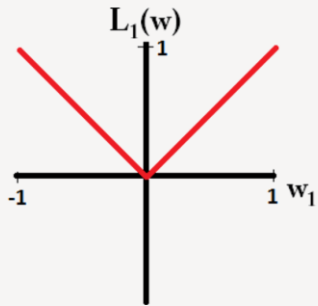
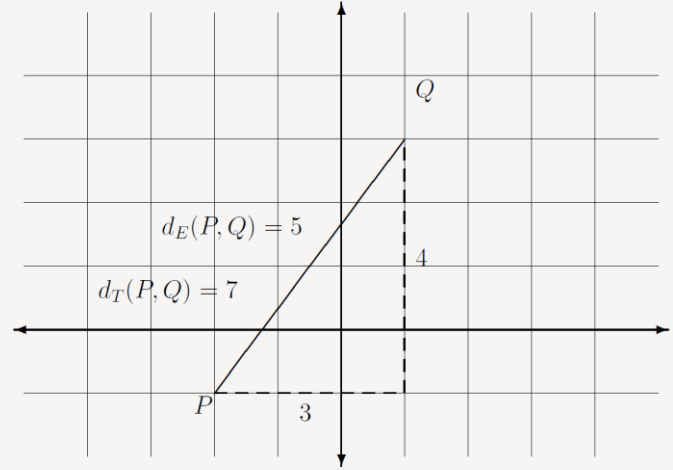


Square l_2 norm



l_1 norm

L1 and L2 Comparisons



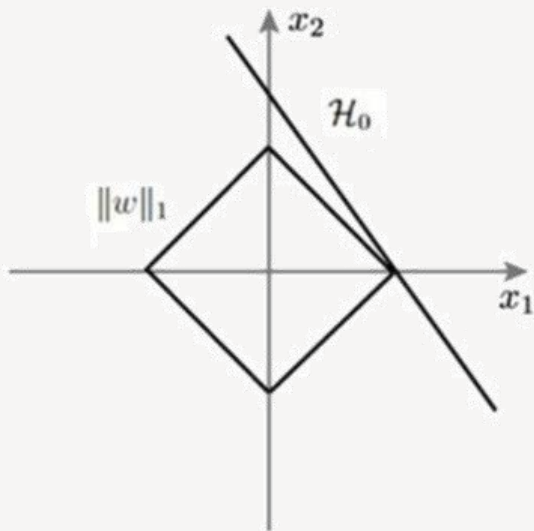
L1 and L2 Comparisons

- ❑ **Robustness** is defined as resistance to outliers in a dataset. The more able a model is to ignore extreme values in the data, the more robust it is.
- ❑ **Stability** is **defined** as resistance to horizontal adjustments. This is the perpendicular opposite of robustness.
- ❑ **Computational difficulty**
- ❑ **Sparsity**

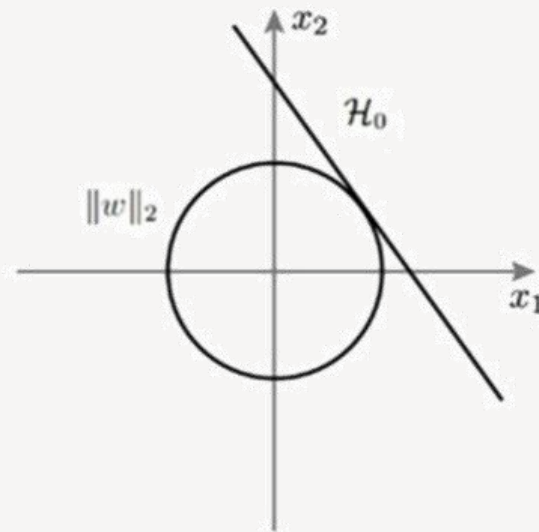


Why is l_1 supposed to lead to sparsity than l_2 ?

$$\min_x \|x\|_{1 \text{ or } 2}, \\ \text{subject to } Ax = b$$



l_1 regularization



l_2 regularization

03

∞ - norm



Vector Norms

□ ∞ -norm(l_∞)(max norm):


$$l_\infty = \max(|x_1|, |x_2|, \dots, |x_n|)$$

- What is the shape of $|x|_\infty = 1$?
- Properties?



04

$\frac{1}{2}$ -norm

Vector Norms

□ $\frac{1}{2}$ -norm($l_{\frac{1}{2}}$)

□ What is the shape of $|x|_{\frac{1}{2}} = 1$?

□ Properties?



Vector Norms

□ 0-norm(l_0):

$$\|x\|_0 = \lim_{\alpha \rightarrow 0^+} \|x\|_\alpha = \left(\sum_{k=1}^n |x|^\alpha \right)^{\frac{1}{\alpha}} = \sum_{k=1}^n 1_{(0,\infty)}(|x|)$$

□ 0-norm, defined as **the number of non-zero elements in a vector**, is an ideal quantity for feature selection. However, minimization of 0-norm is generally regarded as a combinatorially difficult optimization

$$\|x\|_0 = \sum_{x_i \neq 0} 1$$

Vector Norms

- ❑ Is 0-norm a valid norm?
- ❑ What is the shape of $\|x\|_0 = 1$?



Example

- ❑ l_0 distance between $(0, 0)$ and $(0, 5)$?
- ❑ l_0 distance between $(1, 1)$ and $(2, 2)$?
- ❑ (username, password)



Class Activity

- ☐ l_0 distance between $(0, 0)$ and $(0, 5)$?
- ☐ l_0 distance between $(1, 1)$ and $(2, 2)$?
- ☐ $(\text{username}, \text{password})$



Or go to the below link
<https://forms.gle/xFHSDKJDq1KoL4Kx6>

Timer: (2:30 minutes)

Vector Norms

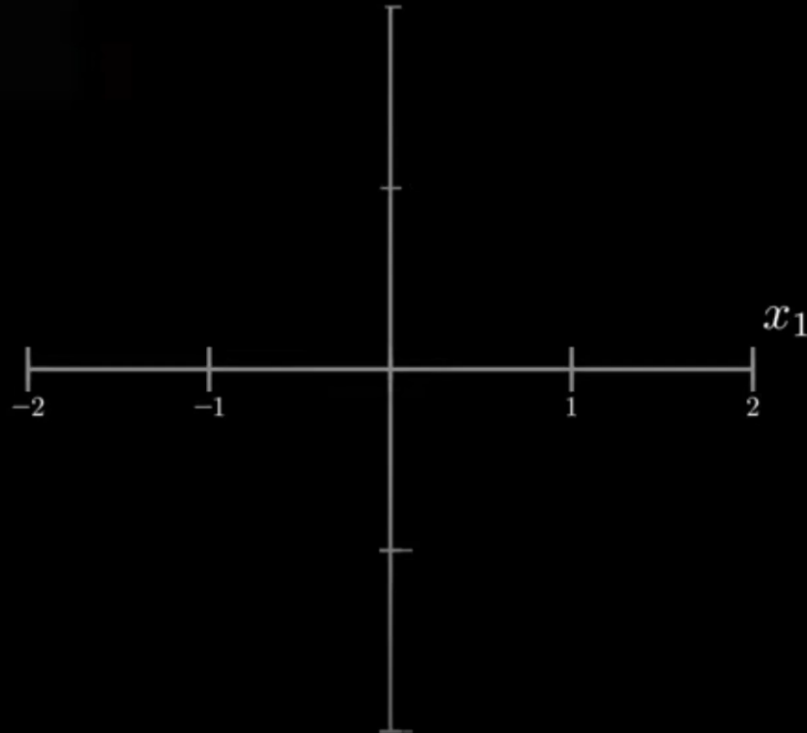
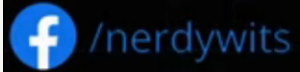
Examples

- l_0 distance between $(0, 0)$ and $(0, 5)$?
- l_0 distance between $(1, 1)$ and $(2, 2)$?
- $(\text{username}, \text{password})$

Solution

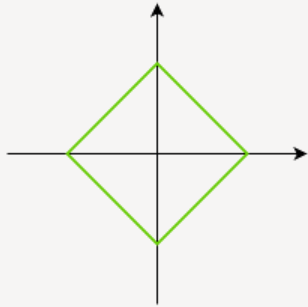
- 1
- 2
- When l_0 is 0, then we can infer that username and password is a match and we can authenticate the user.

Vector Norms Shapes

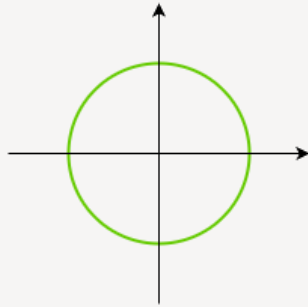


Norms and Convexity

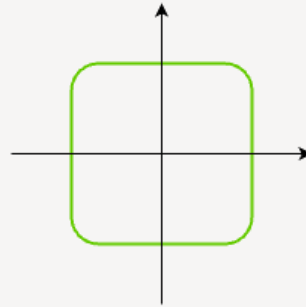
□ For $p \geq 1$, l_p norm is convex



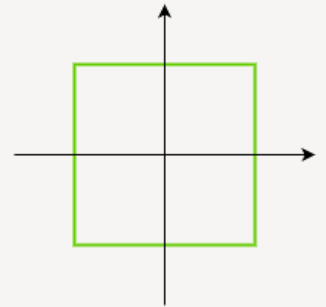
$$\|x\|_1 = 1$$



$$\|x\|_2 = 1$$



$$\|x\|_p = 1$$



$$\|x\|_\infty = 1$$

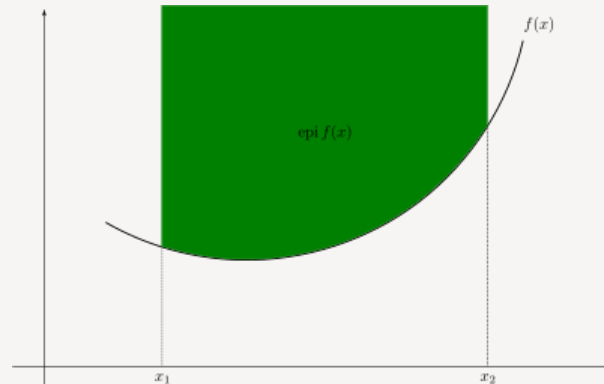


Convex Function

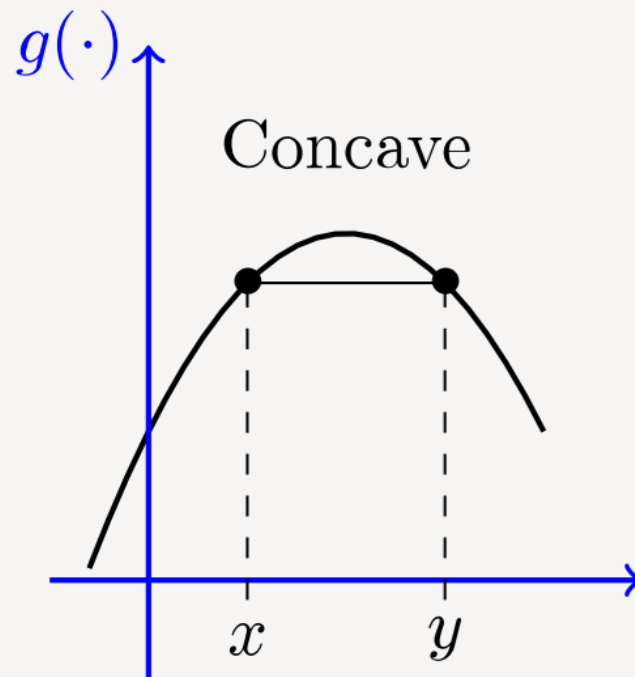
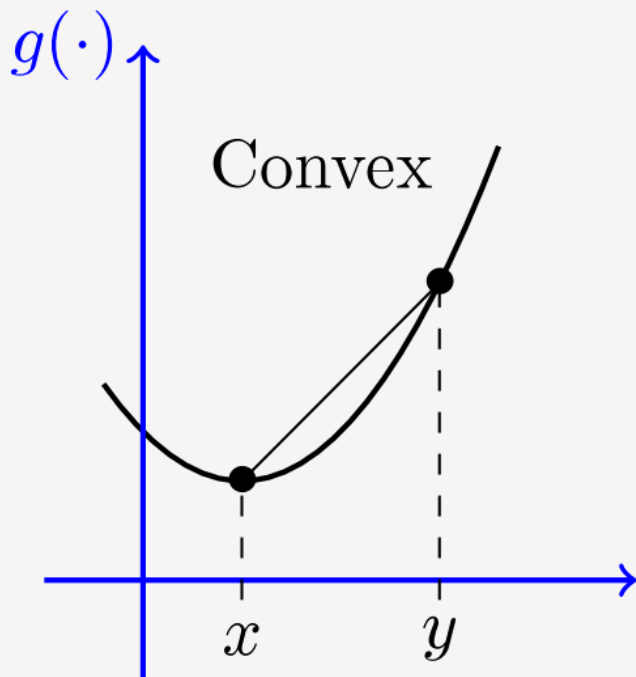
- A function is convex iff its epigraph is a convex set.
- Epigraph or supergraph

$$\text{epi} f = \{(x, \mu) : x \in \mathbb{R}^n, \mu \in \mathbb{R}, \mu \geq f(x)\} \subseteq \mathbb{R}^{n+1}$$

$$f((1-\theta)x^{(0)} + \theta x^{(1)}) \leq (1-\theta)f(x^{(0)}) + \theta f(x^{(1)}), \quad \forall \theta \in [0, 1]$$

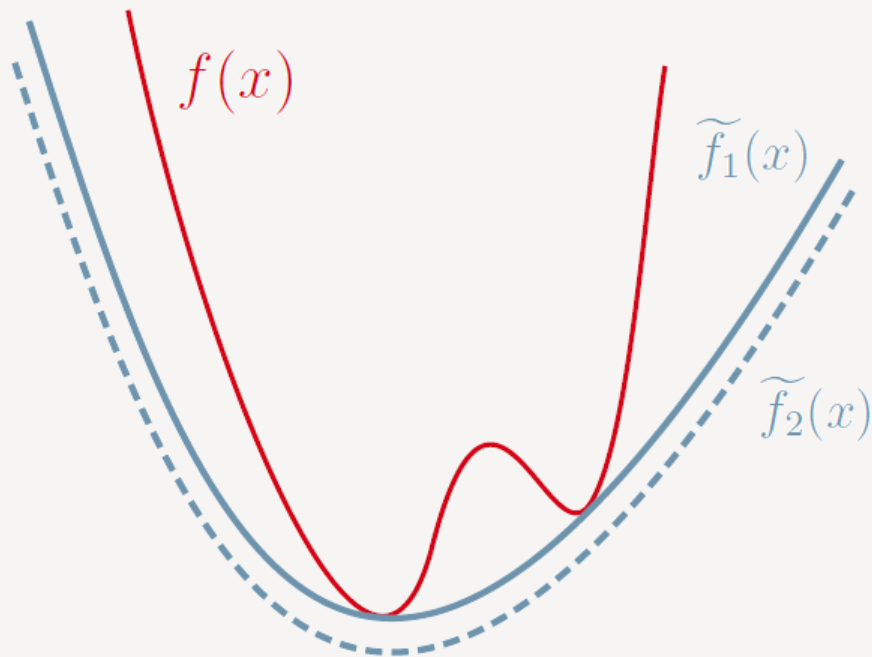


Convex and Concave Function



second derivative is nonnegative on its entire domain

Convex Relaxation

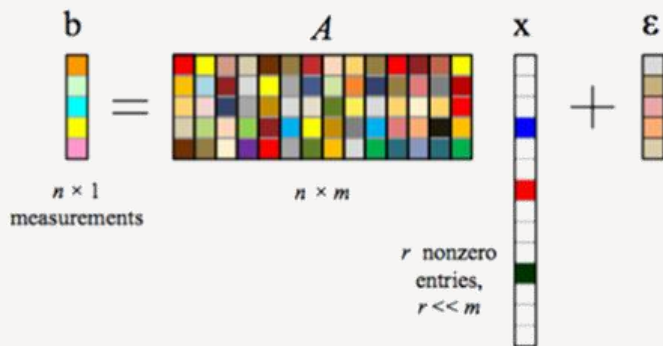


Sparse Applications

- **Alternative viewpoint:** We try to find the sparsest solution which explains our noisy measurements

$$\min_x \|x\|_0, \quad \text{subject to } \|Ax - b\|_2 < \epsilon$$

- Here, the ℓ_0 -norm is a shorthand notation for counting the number of non-zero elements in x .



Sparse Solution

- l_0 optimization is np-hard.
- Convex relaxation for solving the problem.


$$\min_1 \|x\|_1$$

$$\text{subject to } \|Ax - b\|_2 < \epsilon$$

$$\min_1 \|x\|_0$$

$$\text{subject to } \|Ax - b\|_2 < \epsilon$$




L1-L2 norm inequality

Theorem

For all $x \in \mathbb{R}^d$:

$$\|x\|_2 \leq \|x\|_1 \leq \sqrt{d} \|x\|_2$$

Proof

$$\sum_i |x_i| \sum_i |x_i| = \sum_i x_i^2 + \sum_{i \neq j} |x_i| |x_j|$$

Max norm inequality

Theorem

For all $x \in \mathbb{R}^d$:

$$\begin{aligned} \|x\|_{\infty} &\leq \|x\|_1 \leq d \|x\|_{\infty} \\ \|x\|_{\infty} &\leq \|x\|_2 \leq \sqrt{d} \|x\|_{\infty} \end{aligned}$$

Proof

Conclusion

- By a normed linear space (briefly normed space) is meant a real or complex vector space E in which every vector x is associated with a real number $|x|$, called its absolute value or norm, in such a manner **that the properties** (a') – (c') holds. That is, for any vectors $x, y \in E$ and scalar α we have:

- i. $|x| \geq 0$
- ii. $|x| = 0$ iff $x = \vec{0}$
- iii. $|\alpha x| = |\alpha||x|$
- iv. $|x + y| \leq |x| + |y|$

Inner product and norm

Solution

Take any inner product $\langle \cdot, \cdot \rangle$ and define $f(x) = \sqrt{\langle x, x \rangle}$. Then f is a norm.



Proof

Examples

Every inner product gives rise to a norm, but not every norm comes from an inner product. (Think about norm 2 and norm max)



Entry-wise matrix norms

Definition

$$\|A\|_{p,p} = \|vec(A)\|_p = \left(\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^p \right)^{\frac{1}{p}}$$

Special Cases

- ❑ Frobenius (Euclidian, Hilbert Schmidt) norm: ($p = 2$)

$$\|A\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2 \right)^{\frac{1}{2}} = \sqrt{\text{trace}(A^*A)}$$

- ❑ Max norm ($p = \infty$)

$$\|A\|_{\max} = \max_{ij} |a_{ij}|$$

- ❑ Sum-absolute-value norm

$$\|A\|_{sav} = \sum_{i,j} |A_{i,j}|$$

Frobenius (Euclidian, Hilbert Schmidt) norm

Theorem

□ Invariant under rotations (**unitary operations = orthogonal matrices**)

$$\begin{aligned}\|A\|_F &= \|AU\|_F = \|UA\|_F \\ \|A + B\|_F^2 &= \|A\|_F^2 + \|B\|_F^2 + 2\langle A, B \rangle \\ \|A^*A\|_F &= \|AA^*\|_F \leq \|A\|_F^2\end{aligned}$$

$$\|A\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2 \right)^{\frac{1}{2}} = \sqrt{\text{trace}(A^*A)}$$

Frobenius (Euclidian norm)

Theorem

Let $b_1, b_2, \dots, b_n$ denote the columns of B . Then

$$\|AB\|_{HS}^2 = \sum_{i=1}^n \|Ab_i\|^2 \leq \sum_{i=1}^n \|A\|^2 \|b_i\|^2 = \|A\|^2 \|B\|_{HS}^2$$

Using Cauchy-Schawrtz Inequality

Matrix norms induced by vector norms

Definition

$$\|A\|_p = \max_{\vec{x} \neq \vec{0}} \frac{\|A\vec{x}\|_p}{\|\vec{x}\|_p} = \max_{\|\vec{x}\|_p=1} \|A\vec{x}\|_p$$

Theorem

1. Show the above formula is a valid norm.
2. $\|Ax\| \leq \|A\| \|x\|$ for all vectors $\|x\|$
3. For all matrices A, B : $\|AB\| \leq \|A\| \|B\|$

Matrix norms induced by vector norms

Definition

- The norm of a matrix is a real number which is a measure of the magnitude of the matrix.
- Norm 1:

$$\|A\|_1 = \max_{1 \leq j \leq n} \left(\sum_{i=1}^n |a_{ij}| \right)$$

- Norm max:

$$\|A\|_\infty = \max_{1 \leq i \leq n} \left(\sum_{j=1}^n |a_{ij}| \right)$$

Example

$$B = \begin{bmatrix} 5 & -4 & 2 \\ -1 & 2 & 3 \\ -2 & 1 & 0 \end{bmatrix}$$

SVD and Norm

- One common definition for the norm of a matrix is the Frobenius norm:

$$\|A\|_F^2 = \sum_{i=1:m} \sum_{j=1:n} a_{ij}^2$$

- Frobenius norm can be computed from SVD

$$\|A\|_F^2 = \sum_{i=1:p} \Sigma_i^2 \quad \text{where } p = \min(n, m)$$

- So, changes to a matrix can be evaluated by looking at changes to singular values.

SVD and Norm

Theorem

1) Orthogonal matrices, they preserve the Euclidean norm

2) $\|A\|_2 = \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sigma_1$

Norms Compare

The 2-norm (spectral norm) of a matrix is the greatest distortion of the unit circle/sphere/hyper-sphere. It corresponds to the largest singular value (or |eigenvalue| if the matrix is symmetric/hermitian).

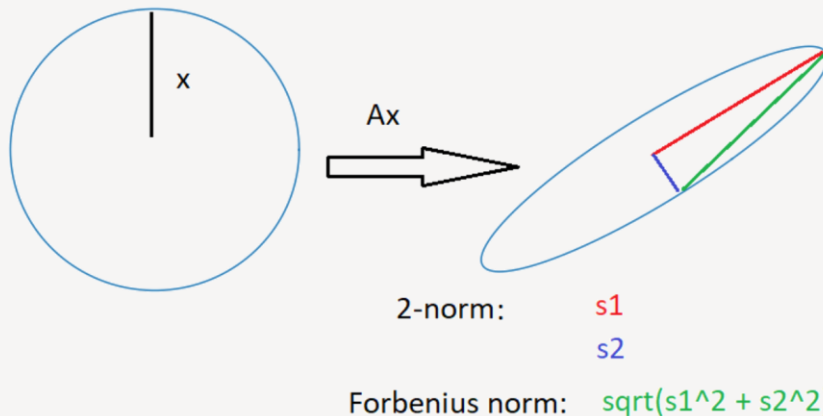
The Forbenius norm is the "diagonal" between all the singular values.

i.e.

$$\|A\|_2 = s_1, \quad \|A\|_F = \sqrt{s_1^2 + s_2^2 + \dots + s_r^2}$$

(r being the rank of A).

Here's a 2D version of it: x is any vector on the unit circle. Ax is the deformation of all those vectors. The length of the red line is the 2-norm (biggest singular value). And the length of the green line is the Forbenius norm (diagonal).



References

- ❑ Linear Algebra and Its Applications, David C. Lay
- ❑ Introduction to Applied Linear Algebra Vectors, Matrices, and Least Squares
- ❑ <https://www.youtube.com/watch?v=76B5cMEZA4Y>

